Final Project Document
“Wijhaty Application”

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Main Objective

Wijhaty is designed to make course booking simple and convenient for users. The system lets users browse available courses, choose one, enter their details, and confirm their booking easily. It also ensures that all inputs are valid, saves booking details securely, and provides a clear summary for the user. The goal is to offer a straightforward and reliable experience that anyone can use without difficulty.

General Functionality of the Project:

1. **Main Window**: The application begins with a welcome message and displays the app's logo.
2. **Option Window**: After the main window, the user is presented with options to select the specialization.
3. **Course Selection (AICourseWindow and DSCoursesWindow)**: Based on the user's choice from the Option window, they are directed to either the *AICourseWindow* or *DSCoursesWindow*. In these windows, they can choose from a list of available courses that match their preferences.
4. **Booking Window**: Once the user selects a course, they click the "Book" button, which navigates them to the *BookWindow*. Here, they are prompted to enter their personal information .
5. **File Handling**: In the *BookWindow*, the user's booking details are saved to a file for future reference. This ensures the information is stored securely and can be retrieved later.
6. **Booking Information Display**: In the *BookingInfoDisplay* window, the saved booking details are read from the file and displayed to the user, providing a summary of their booking.

Layout and Design

The application uses BorderLayout, GridLayout, and FlowLayout for organizing components, including JLabels for instructions, JTextFields for input, JButtons for actions, and a JTextArea to display booking information

Component Orientation:

All components are set to right-to-left to accommodate Arabic content.

Colors and Fonts

- The window uses a white background (Color.WHITE) with turquoise blue `(new Color(47, 140, 163))` for buttons and labels.
- Arial, bold is used for buttons and labels, while Times New Roman is used for the booking information area.
-

File Handling:

Booking information is saved to a text file (booking_info.txt) in BookWindow and displayed in BookingInfoDisplay window, showing the last user entries.

Main Window

تطبيق وجهتي
ملف حول
مرحبا بكم في تطبيق وجهتي
وجهتي
التالي
ملف حول
إغلاق
ملف حول
عن التطبيق
اتصل بنا
Message
هذا اصدار تجريبي لتطبيق وجهتي.
OK
Message
تواصل معنا على: support@wuajhati.com.
OK

Functionality:

The MainWindow serves as the initial screen of the *Wijhaty* application. It welcomes users with a message and logo, and provides a button that leads to the next window (OptionWindow). Additionally, the window includes a menu bar with options for closing the application or showing "About Us" and "Contact Us" information.

Components:

1. **JPanel**: The primary container that holds the components in the window, organized using BorderLayout.
2. **JLabel**: Used for displaying the welcome message and logo.
3. **ImageIcon**: Used to display the app logo on the window.
4. **JButton**: The "التالي" button that lets the user proceed to the next window (OptionWindow).
5. **JMenuBar**: Holds the menus for the app, including options like "File" and "About".
6. **JMenu**:
    - "ملف" menu for file-related options (close application).
    - "حول" menu for app details and contact info.
7. **JMenuItem**: Menu items like "عن التطبيق", "إغلاق", and "اتصل بنا" with their corresponding actions.

Layout:

- The **MainWindow** uses **BorderLayout** to arrange components:
    - **Center**: Displays the app logo.
    - **North**: Displays the welcome message.
    - **South**: Contains the "التالي" button.
- **MenuBar** is placed at the top to manage user actions (File and About).

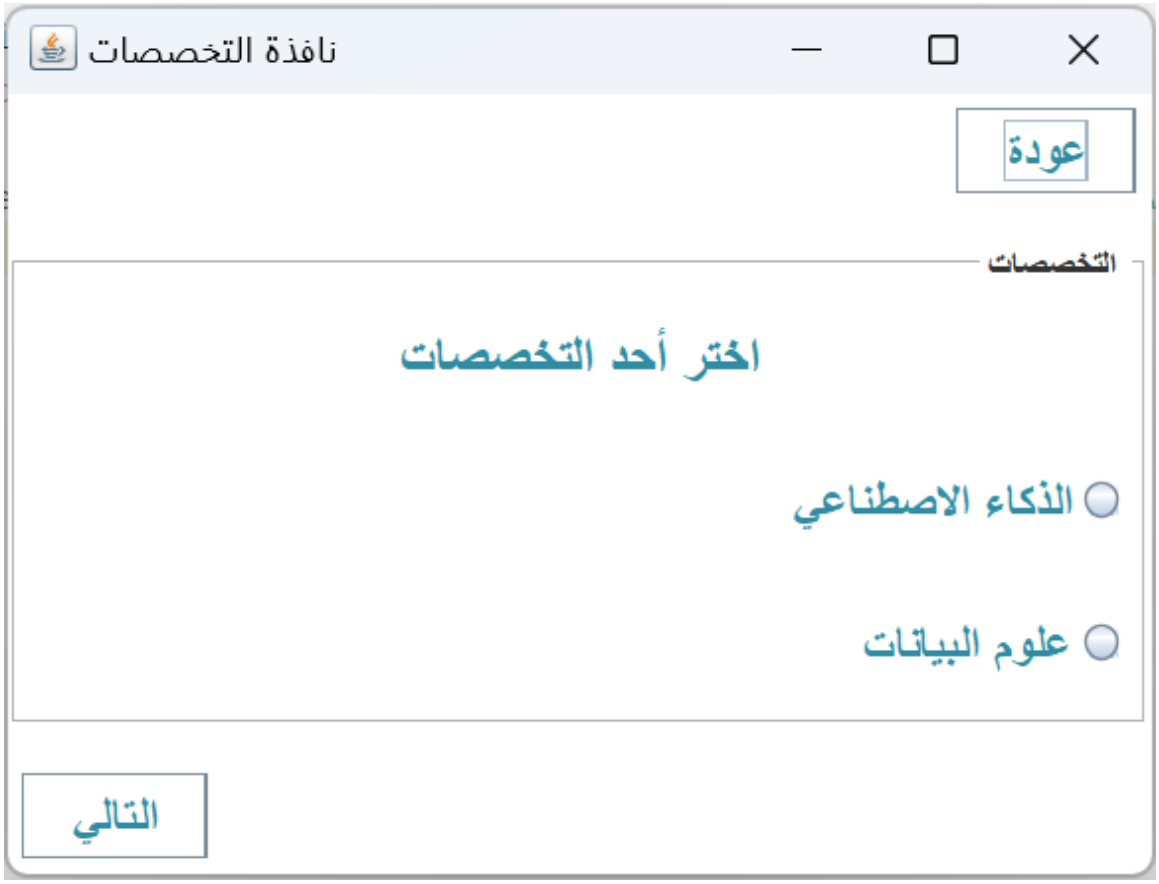
Font:

- Arial, bold, size 20 for the welcome message and button labels.

Colors:

- **Background Color**: The window's background is the default color.
- **Text Color**: Welcome message, button text, and menu items use turquoise blue `(new Color(47, 140, 163))`.
- **Button Color**: The **"التالي"** button has a turquoise blue background `(new Color(69, 191, 212))` with white text.

OptionWindow



Functionality:

- The OptionWindow allows users to choose between two specialized courses: Artificial Intelligence (AI) and Data Science (DS).
- The window consists of radio buttons for course selection, with a "Back" button for navigation to the main window and a "Next" button to proceed to the appropriate course details page, based on the user's choice.
- If no course is selected, the system prompts the user to select a course before proceeding.

Components:

- **JRadioButton:** The AI and DS courses are represented using JRadioButton components, which allow users to select one option.

- **JButton**: The Next and Back buttons allow for navigation between windows.
- **JLabel**: A JLabel displays the title "اختر أحد التخصصات" , indicating the purpose of the window.
- **ButtonGroup**: ensures that only one course option (either AI or DS) can be selected at a time.
- **BorderFactory**: A titled border created using `BorderFactory.createTitledBorder` is applied around the course selection panel, clearly labeling it as "التخصصات"
- **JPanel**:
    - **panel1**: This panel contains the Back button and is aligned to the right using FlowLayout.
    - **panel2**: This panel contains the course selection radio buttons (AI and DS) and the title label, arranged in a GridLayout to stack them vertically. It also features a titled border for visual clarity.
    - **panel3**: This panel contains the Next button, aligned to the left using FlowLayout.

Layout:

- **BorderLayout**:
    - **panel1** (Back button) is placed in the **North**.
    - **panel2** (Course selection) is placed in the **Center**.
    - **panel3** (Next button) is placed in the **South**.
- **FlowLayout:** Used to set the alignment and positioning of components. The "Back" button is aligned to the top right, while the "Exit" or "Next" button is centered at the bottom, ensuring a clean and user-friendly layout.
- **GridLayout**: GridLayout arranges the course selection radio buttons and the title label in the center panel, ensuring they are displayed in a vertical column.

Font and Colors:

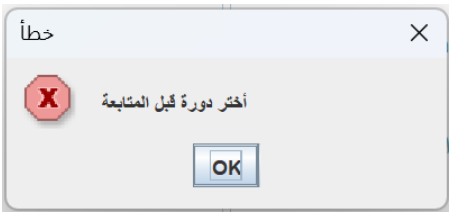
- **Font**: **Arial**, **bold** for button, labels and the title. With size 18 for title and size 16 for buttons and radio buttons.

- **Colors**:
    - **Background Color**: White.
    - **Text Color**: turquoise blue color used for labels, buttons, and radio buttons represented by `new Color(47, 140, 163)`.
    - **Button Color**: white background and turquoise blue text.

Event Handling:

- **NextButtonListener**: The Next button triggers an action listener that checks if a course is selected. If so, it navigates the user to the corresponding course window (either AICoursesWindow or DSCoursesWindow). If no course is selected, it prompts the user to make a choice.
- **BackButtonListener**: The Back button allows the user to return to the MainWindow.

AICoursesWindow



Functionality:

- The AICoursesWindow allows users to select from a list of artificial intelligence courses, categorized into Beginner and Advanced levels.
- The window includes a Back button to return to the previous window (the OptionWindow) and a Book button to proceed with booking once a course is selected.

- The course options are displayed using radio buttons, and the user can select one course from each category.
- If no course is selected, an error message prompts the user to choose a course before proceeding with booking.

Components:

- **JRadioButton**: The course options are represented by JRadioButton components, with separate buttons for beginner and advanced courses.
    - **Beginner Courses**: "برمجة بايثون" , "مقدمة في الذكاء الاصطناعي" , and " مقدمة في تعلم الآلة"
    - **Advanced Courses**: "معالجة اللغات الطبيعية" , "التعلم العميق" , and " الرؤية الحاسوبية".
- **JButton**: Back button navigates the user to the previous window and a Book button to proceed with booking.
- **ButtonGroup**: A ButtonGroup is used to ensure that only one course can be selected from each group of beginner and advanced courses.
- **JLabel**: A JLabel is used to display the title "قائمة دورات الذكاء الاصطناعي" , which is centered at the top of the window.
- **BorderFactory**: The BorderFactory is used to create titled borders around the Beginner Courses and Advanced Courses panels.
- **JPanel**: Panels are used to organize and structure the window layout:
    - **topPanel**: This panel includes the Back button and the title label. It uses BorderLayout to position the components in the East (for the Back button) and Center (for the title).
    - **beginnerPanel**: This panel, which is arranged with a GridLayout, contains the beginner courses and is bordered with a titled label "دورات للمبتدئين".
    - **advancedPanel**: Similarly, this panel is arranged with a GridLayout and contains the advanced courses, bordered with the label "دورات متقدمة".
    - **mainPanel**: The mainPanel organizes the beginnerPanel and advancedPanel side by side using a GridLayout with 1 row and 2 columns.

Layout:

- **BorderLayout**: The overall layout of the window uses BorderLayout, with the topPanel positioned at the North, the mainPanel (containing both course panels) at the Center, and the Book button at the South. It is also used in topPanel to align Back button to the East and title label to the center
- **GridLayout**: The beginnerPanel and advancedPanel are arranged using a GridLayout

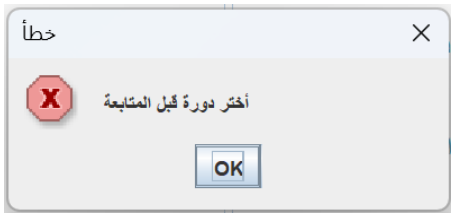
Font and Colors:

- **Font**: Arial, bold, size 24 for title label and 16 for other components.
- **Colors**:
  - **Background Color**: White.
  - **Text Color**: turquoise blue `(new Color(47, 140, 163))`
  - **Button Color**: The Back button has a white background with turquoise blue text, and the Book button has a light blue background `(new Color(69, 191, 212))` with white text.

Event Handling:

- **BackButtonListener**: The **Back** button triggers an action listener that allows the user to return to the **OptionWindow** when clicked.
- **BookButtonListener**: The Book button triggers an action listener that checks if a course is selected. If a course is selected, the user is directed to the BookWindow to enter their details. If no course is selected, the user is prompted with an error message to select a course before proceeding.

DSCoursesWindow



Functionality:

- The DSCoursesWindow allows users to select from a list of Data Science courses, categorized into Beginner and Advanced levels.
- The window includes a Back button to return to the previous window (OptionWindow) and a Book button to proceed with booking once a course is selected.
- The course options are displayed using JRadioButton components, and the user can select one course from each category.

- If no course is selected, an error message prompts the user to choose a course before proceeding with booking.

Components:

- **JRadioButton**: The course options are represented by JRadioButton components, with separate buttons for beginner and advanced courses.
    - **Beginner Courses**: "مقدمة في علوم البيانات" , "تحليل البيانات", and "نمذجة البيانات".
    - **Advanced Courses**: "تعلم الآلة", "ذكاء الأعمال" , and "البيانات الضخمة" .
- **JButton**: The window includes a Back button for navigation back to the OptionWindow and a Book button to proceed with booking.
- **ButtonGroup**: A ButtonGroup is used to ensure that only one course can be selected from each group of beginner and advanced courses.
- **JLabel**: A JLabel is used to display the title "قائمة دورات علوم البيانات" , centered at the top of the window.
- **BorderFactory**: The BorderFactory is used to create titled borders around the Beginner Courses and Advanced Courses panels.
- **JPanel**: Panels are used to organize and structure the window layout:
    - **topPanel**: This panel includes the Back button and the title label. It uses BorderLayout to position the components in the East (for the Back button) and Center (for the title).
    - **beginnerPanel**: This panel, which is arranged with a GridLayout, contains the beginner courses and is bordered with a titled label "دورات للمبتدئين" (Beginner Courses).
    - **advancedPanel**: Similarly, this panel is arranged with a GridLayout and contains the advanced courses, bordered with the label "دورات متقدمة" (Advanced Courses).
    - **mainPanel**: The mainPanel organizes the beginnerPanel and advancedPanel side by side using a GridLayout with 1 row and 2 columns.

Layout:

- **BorderLayout**: The overall layout of the window uses BorderLayout, with the topPanel positioned at the North, the mainPanel (containing both course panels) at the Center, and the Book button at the South.

- **GridLayout**: The beginnerPanel and advancedPanel are arranged using GridLayout to ensure that each course is displayed in a vertical stack.
- **FlowLayout**: The Back button is aligned to the East of the topPanel, while the title label is centered in the topPanel using BorderLayout.

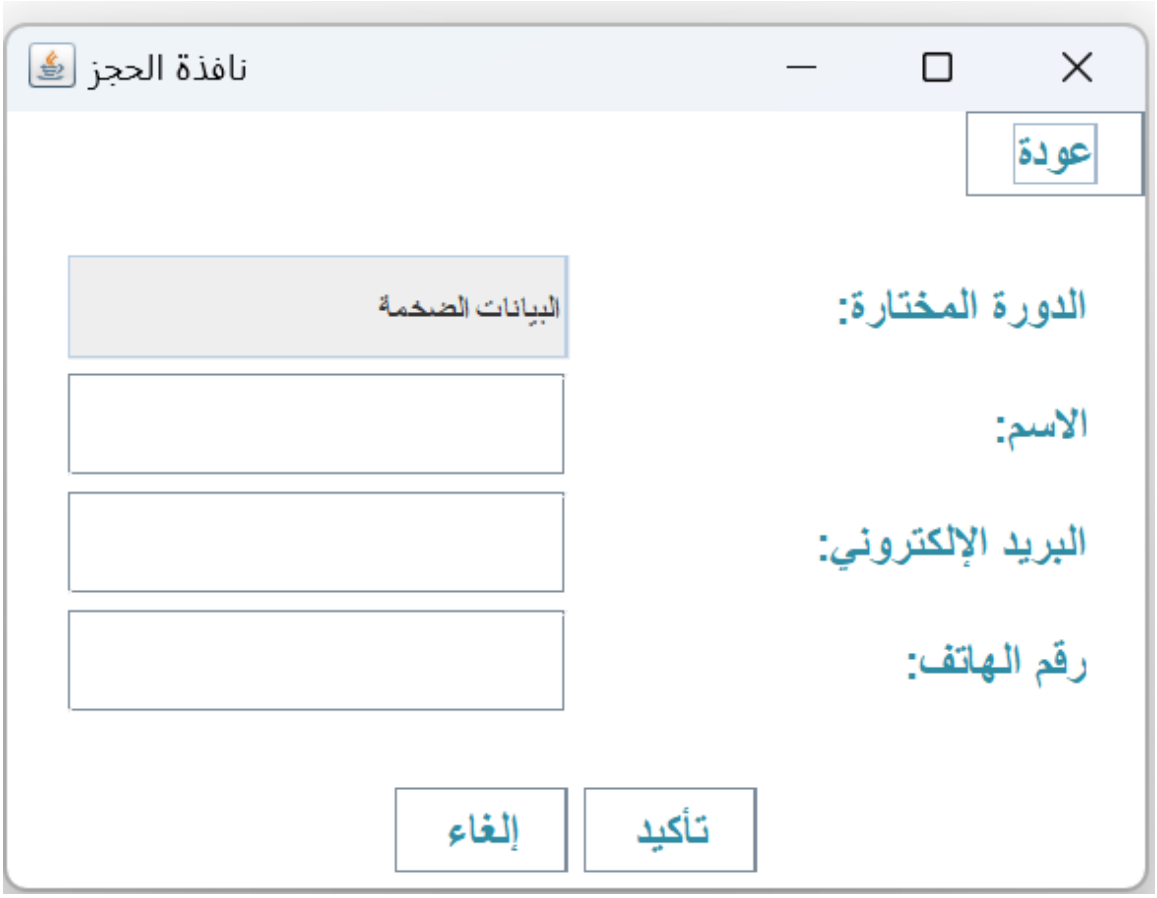
Font and Colors:

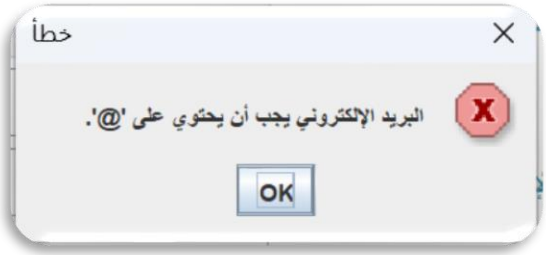
- **Font**: Arial, bold, size 24 for title label and 16 for other component
- **Colors**:
    - **Background Color**: White.
    - **Text Color**: **turquoise blue** `(new Color(47, 140, 163))`
    - **Button Color**: The Back button has a white background with turquoise blue text, and the Book button has a light blue background `(new Color(69, 191, 212))` with white text.

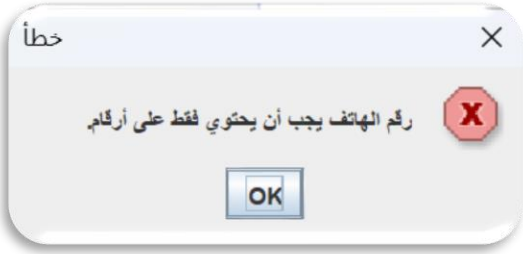
Event Handling:

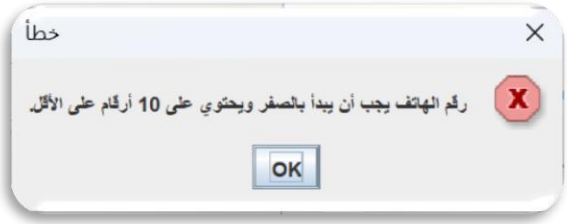
- **BackButtonListener**: The Back button triggers an action listener that allows the user to return to the OptionWindow when clicked.
- **BookButtonListener**: The Book button triggers an action listener that checks if a course is selected. If a course is selected, the user is directed to the BookWindow to enter their details. If no course is selected, the user is prompted with an error message to select a course before proceeding.

BookWindow


نافذة الحجز
عودة
الدورة المختارة:
البيانات الضخمة
الاسم:
البريد الإلكتروني:
رقم الهاتف:
تأكيد
إلغاء


خطأ
البريد الإلكتروني يجب أن يحتوي على '@'.
OK


خطأ
رقم الهاتف يجب أن يحتوي فقط على أرقام.
OK


خطأ
رقم الهاتف يجب أن يبدأ بالصفر ويحتوي على 10 أرقام على الأقل.
OK

Functionality:

- The **BookWindow** is used to collect booking information for a selected course, including the user's name, email, and phone number.
- The **selectedCourse** is displayed in a non-editable field, and the user must fill out the remaining fields: name, email, and phone.
- The window contains two main buttons:
  - **Back Button**: Navigates the user back to the previous window (calling window).
  - **Confirm Button**: Saves the booking details if valid and displays a success message. The information is written in a file (booking_info.txt).
  - **Cancel Button**: Exits the application when clicked.
- The input fields are validated for proper email format and phone number requirements before proceeding with saving the data.

Components:

- **JTextField**: Four text fields are used to collect data:
  - **courseField**: Displays the selected course (non-editable).
  - **nameField**: For entering the user's name.
  - **emailField**: For entering the user's email address.
  - **phoneField**: For entering the user's phone number.
- **JButton**:
  - **Back Button**: Returns to the calling window.
  - **Confirm Button**: Saves the input data if the validation passes.
  - **Cancel Button**: Closes the application.
- **JLabel**: Labels are created via createLabel() method and include:
  - "الدورة المختارة" (Selected Course), "الاسم" (Name), "البريد الإلكتروني" (Email), "رقم الهاتف" (Phone Number)
- **JPanel**:

- **mainPanel**: The primary container for the window layout, with a BorderLayout to manage the positioning of elements.
- **topPanel**: Holds the Back button, positioned in the east.
- **contactPanel**: A GridLayout panel that holds the text fields and their respective labels.
- **buttonPanel**: A panel at the bottom with Confirm and Cancel buttons arranged using FlowLayout.

Layout:

- **BorderLayout**:
  - The **topPanel** is placed at the North to hold the Back button.
  - The **mainPanel** (containing the contact form) is positioned at the Center.
  - The **buttonPanel** is positioned at the South for the Confirm and Cancel buttons.
- **GridLayout**: The contactPanel uses GridLayout with 4 rows and 2 columns
- **FlowLayout**: The buttonPanel uses FlowLayout with center alignment for positioning the buttons.

Font and Colors:

- **Font**: Arial, bold, size 16 for labels and buttons.
- **Colors**:
  - **Background Color**: White
  - **Text Color**: turquoise blue `(new Color(47, 140, 163))`
  - **Button Color**: The Back, Confirm, and Cancel buttons use a white background with turquoise blue text, ensuring they stand out clearly.

Event Handling:

- **BackButtonListener**: The Back button has an action listener that closes the BookWindow and reopens the callingWindow (the previous window the user came from).

- **ConfirmButtonListener**: The Confirm button listens for clicks and performs the following actions:
    - Collects values from the input fields.
    - Validates the email and phone number format:
        - Checks if the email contains the @ symbol.
        - Verifies the phone number starts with 0 and is at least 10 digits long, containing only numbers.
    - If the input is valid, the booking details are saved to a file (booking_info.txt), and a success message is displayed.
    - After saving, a new **BookingInfoDisplay** window is created.
- **CancelButtonListener**: The Cancel button terminates the program when clicked.

Validation:

- The **validateInput** method ensures the email contains the @ symbol and that the phone number is valid by:
    - Checking the length and format of the phone number (it should start with 0 and contain 10 digits).
    - Verifying that the phone number contains only numeric characters.
- If any validation fails, an error message is shown using `JOptionPane.showMessageDialog()`.

Saving Booking Information:

- The **saveBookingInfo** method writes the user's booking data to a file (booking_info.txt) using a **BufferedWriter**.
    - The data includes the selected course, user’s name, email, and phone number.
    - A success message is shown, and the window is closed.

BookingInfoDisplay

Functionality:

- The BookingInfoDisplay window displays recent booking information to the user.
- It reads the booking details from a file named booking_info.txt.
- The window provides the following actions:
  - A **Back** button to return to the BookWindow.
  - An **Exit** button to close the application.
  - Displays a **"شكرًا لك "** message below the booking details.

Components:

1. **File Reading and Display**:

- Reads the booking_info.txt file using a Scanner.
- Displays the last five lines of the file to show recent booking details.

2. **JTextArea**:
   - Used to display booking information in a scrollable and non-editable text area.
   - Configured for right-to-left orientation to align with the Arabic content.
3. **JButton**:
   - **Back Button**: Returns the user to the BookWindow.
   - **Exit Button**: Terminates the application.
4. **JLabel**:
   - Title: Displays "معلومات حجزك" at the top of the window.
   - Footer: Displays a **"شكرا لك"** message at the bottom.
5. **JScrollPane**:
   - Ensures the JTextArea is scrollable in case the booking information exceeds the visible area.
6. **Panels**:
   - **Top Panel**: Contains the title label and the back button, aligned using FlowLayout.
   - **Button Panel**: Contains the "شكرًا لك" label and the exit button, arranged with BorderLayout.

Layout:

- **BorderLayout**:
  - Top Panel: Contains the title and back button.
  - Center: Displays the booking information in a scrollable area.
  - Bottom Panel: Contains the "Thank You" label and exit button.

Font and Colors:

- **Font**:
  - Title and "Thank You" label: Arial, bold, size 16-18.
  - Booking Information: Times New Roman, bold, size 16.
- **Colors**:
  - Background: White.
  - Buttons:
    - Back Button: Turquoise `(new Color(47, 140, 163))` text on white background.
    - Exit Button: Light blue background with white text.

Event Handling:

1. **BackButtonListener**: Hides the BookingInfoDisplay window and makes the BookWindow visible again. So it prevents the application from terminating, maintaining user progress.
2. **ExitButtonListener**: Exits the application immediately when the "Exit" button is clicked.